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CSC 6303 Week 7

OS examples: Linux, MacOS, and Windows

Systems and Languages Survey - Dr. Paulo Fernandes

Presentation Agenda

Week 7

- Virtual Machines
  a. Principles
  b. VirtualBox
  c. Google Colab
- OS Examples
  a. Linux
  b. MacOS
  c. Windows
- This Week's tasks

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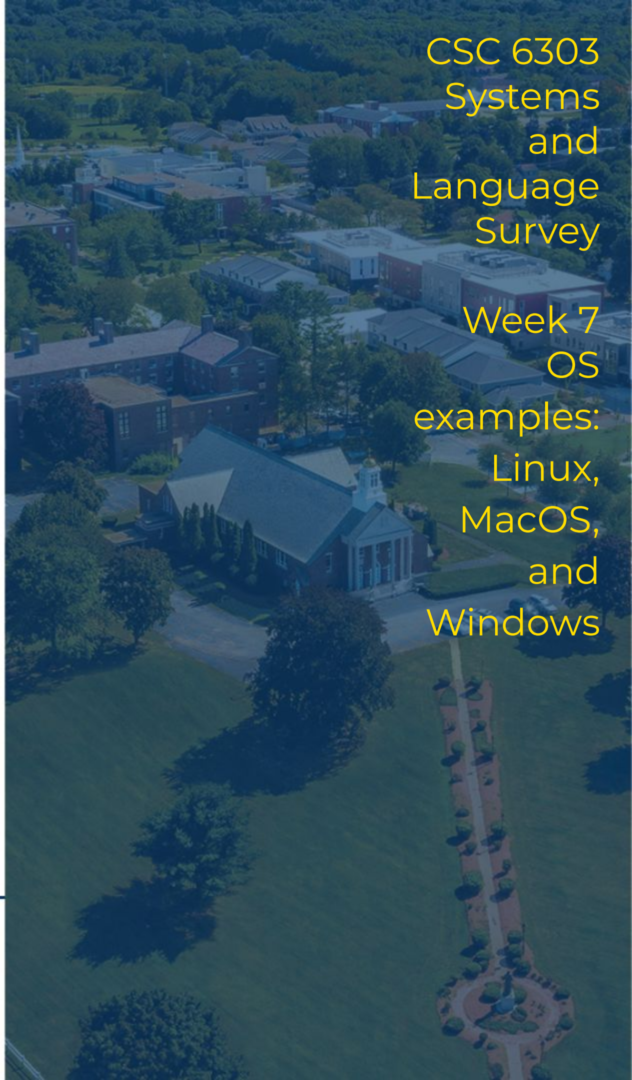


Virtual Machines

All machines are hardware, but all we actually use is software.

There was a time that you bought a machine (hardware) and them you choose the software to run on it.

Then, at some point, we started to buy a machine and having the software, the OS coming together.

Virtual Machine (VM) is an easy way to go back to the past when we could choose the OS for our hardware. We will see VMs by its:

- Principles
- A practical example: VirtualBox

Virtual Machines

While using your OS you have access to computer resources, including the ability to grant access to applications designed for your OS.

However, you cannot grant access to applications that are not designed to your OS because they do not use the same protocol your OS requires.

A virtual machine app provides the translation in order to execute another OS that can then execute applications designed for it. A virtual machine app provides a virtual machine monitor that can support virtual machines running their own OS, both for your hardware or someone else's (the cloud, for example).

solarwinds vmware aws

Wikipedia: Virtual Machine.

Virtual Machines Kinds

- If you have a virtual machine app is running over your hardware and translating the virtual OS execution to your hardware, you have a traditional virtual machine app, as:
  - Oracle's VirtualBox;
  - Parallels;
  - VMware's Desktop products;
- If you have a virtual machine app running over your hardware (maybe through your browser) and relaying your commands to a remote hardware executing the virtual OS, you have a cloud based virtual machine, as:
  - Microsoft's Azure;
  - Amazon's AWS;
  - Paperspace's Gradient;
  - Google's Colab.

Text: Best VM apps.

Oracle's VirtualBox

This virtual machine app is a standard in the free software and it allows you to install and run any of the major OS in your machine.

Once the app installed in your machine (and native OS), you can install virtual machines (assuming you have the OS installer).

Your own native operating system is called ***host OS***, while the OS installed on the virtual machine is called ***guest OS***.

The installations should be pretty straightforward, but depending on several previously configured options, you might face some hiccups. Do not hesitate to search the Internet for help, because if you had a hiccup, others might as well.

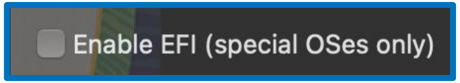
Resource: VirtualBox official page.

Video: Installing Windows 11, Linux, and MacOS VMs.

VirtualBox common hiccups

- Releasing the mouse cursor captured on the Virtual Machine:
  - on a Mac: left command key (left to space bar);
  - on a PC (Windows or Linux): right shift key (right to question mark key);

- Setting up the Virtual Machine resolution:
  - change the screen resolution in the guest OS, VirtualBox will adjust;
- Installing Windows 11 on a MacOS Intel-based:
  - The EFI option usually needs to be turned off;



- Installing MacOS in a PC:
  - The keyboard may need adjust within the VirtualBox (the right most icon in the VM window)

- Pay attention to the main user name and password!

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Be patient ... virtual machines may be extremely slow.

Google Colab

This virtual machine service is one of the options to use virtual machines offered by a commercial provider. Besides the paid options, you can use a free account that is enough to run simple programs.

This machine is capable to see your Google virtual drive to access files you upload there.

By default the ***guest OS*** of a Google Colab virtual machine is a Linux OS.

The usage should be simple if you have a Google account (like your Merrimack account or your personal @gmail.com address). The Google Colab entry page is very well explained, but do not hesitate to search the Internet for help, because if you had a hiccup, others might as well.

Resource: Google Colab entry page.

OS Examples

A Computer Scientist cannot say I don't know an OS, just like a chef cannot say I don't know a kitchen.

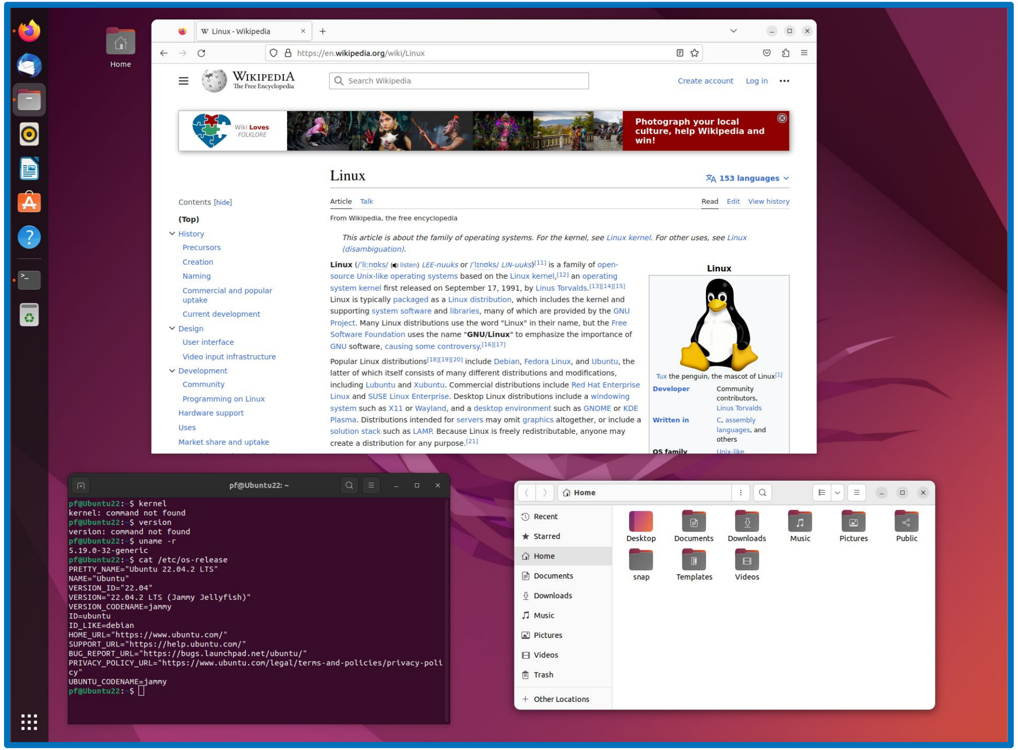
All programming languages have the same power, but clearly some are better suited to some tasks. The same goes for Operating Systems, they all have the same power. Some are lighter, some are prettier. Other than that, just like languages, it is a matter of taste. At least among:

- Linux,
- MacOS, and
- Windows.

Linux

This is not an OS, but a family of open-source OSs, the particular distribution we will see is ***Ubuntu Desktop*** version ***22.04.02***.

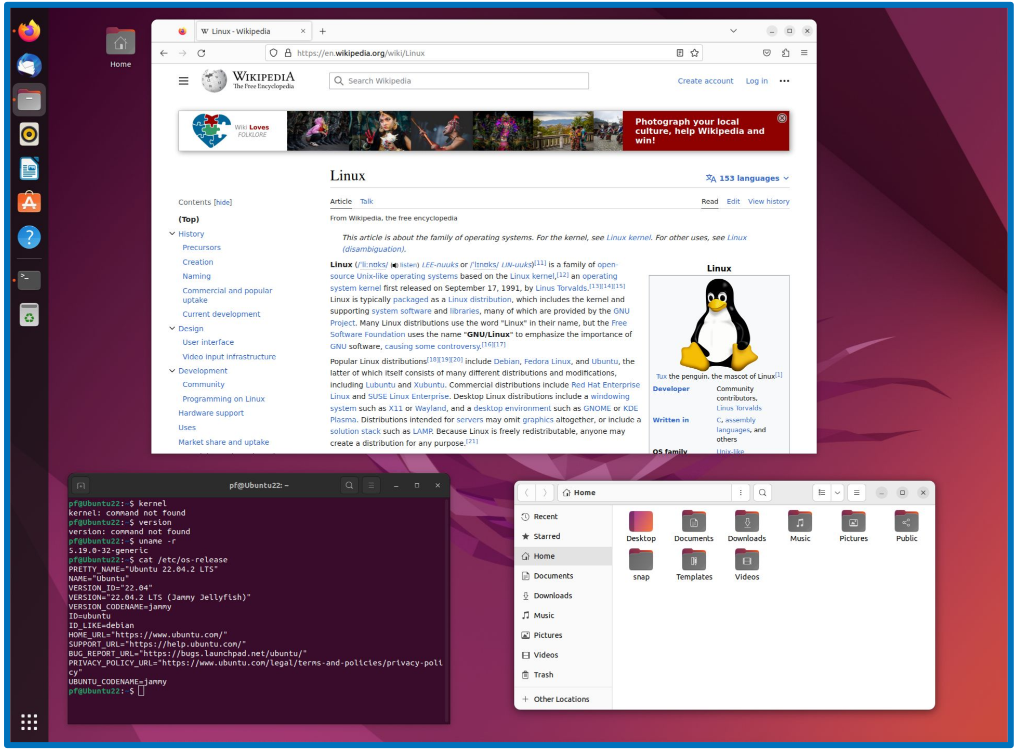
- To open a terminal click at the app menu at the bottom-left corner
- To check the version out type in the terminal ***cat /etc/os-release***



Wikipedia: Linux.

Linux

- The setting are also accessible by the app menu;
- The native browser is ***Firefox*** app;
- The GUI file system navigator is the ***File*** app;
- Usual productivity suite is ***LibreOffice*** apps;
- Try the Mahjongg out!



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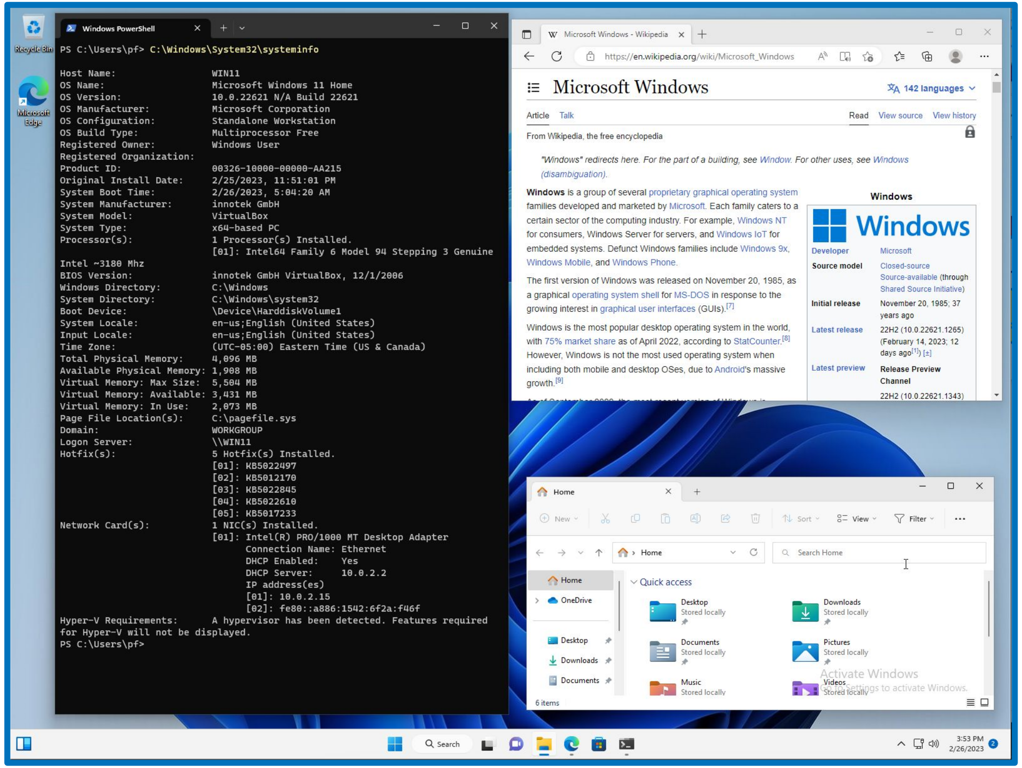
Official page: Linux - Ubuntu.

Windows

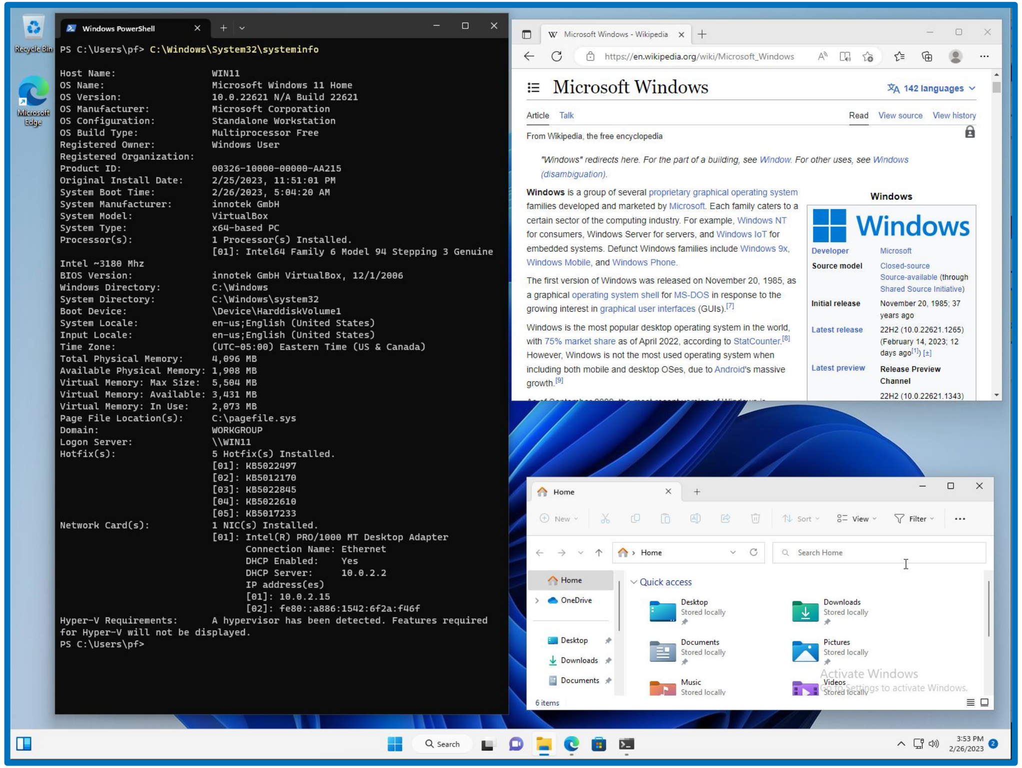
Windows is a group of several proprietary OS that included older versions, and currently two versions of end-user OS's (***Windows 11*** and ***Windows 11 Enterprise***), plus a server version (***Windows Server 2022***) that is considerably different from the other two.



Wikipedia: Windows.

Windows

- To open PowerShell go to the main menu and type in ***PowerShell***
- To check the version out type in PowerShell ***C:\Windows\System32\systeminfo***
- The settings are also at the main menu;
- The native browser is ***Edge*** app;

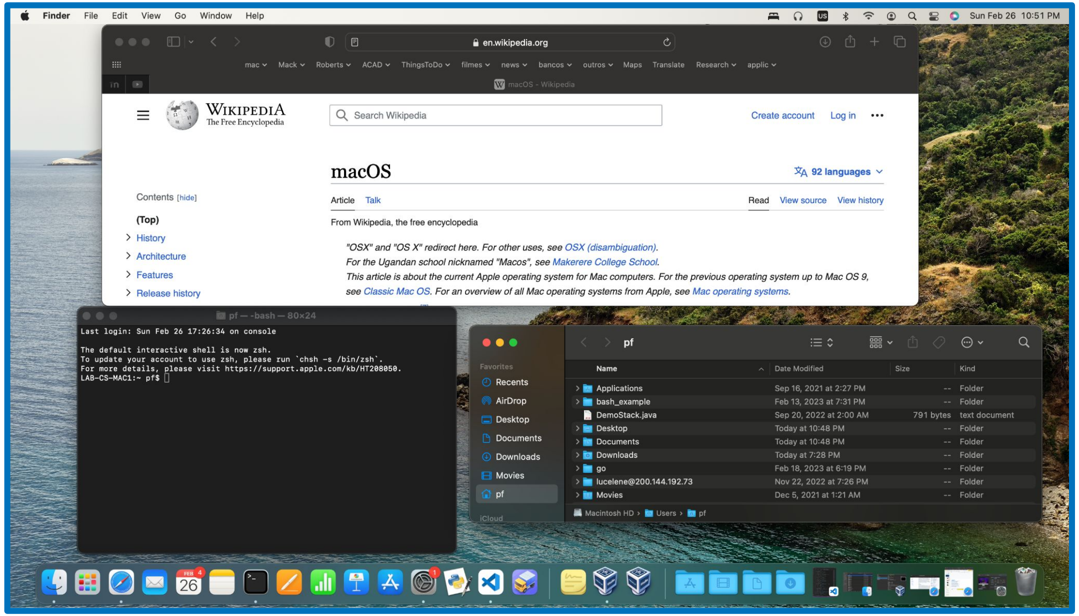


Official page: Windows 11.

MacOS

Mac OS too is not a single OS, but it is used to describe several versions of Apple computers' OSs. Among those, *Mac OS X* is a fully Unix-like OS being launched in 2001.

The current version is ***14.0.0*** (***Sonoma***), but the one shown here is the twice previous one ***12.6.3*** (***Monterrey***).



You can check your version opening a terminal and typing ***sw_vers*** command.

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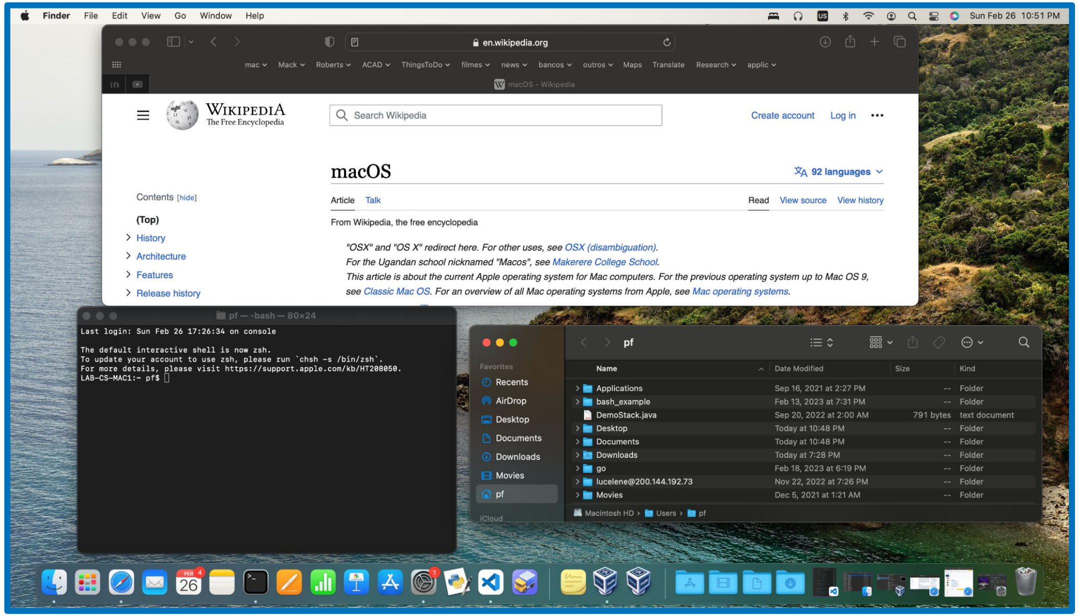
Wikipedia: MacOS.

MacOS

- To open the terminal you can open the ***Applications*** folder, go to ***Utilities*** and chose ***Terminal***;
- The standard browser is the ***Safari*** app and the GUI file system navigator is the ***Finder*** app;
- The settings are also accessible through the ***Applications*** folder.



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Official page: MacOS.

Other OSs

Other OS are currently available and that list includes:

- the specific ones for mobile devices, as iOS, watchOS, Android, Mobile Linux, Windows 11 mobile, etc;
- less popular options (at least so far), as:
  - ChromeOS, the free OS by Google, is available to be installed in desktop/laptop machines, and obviously on Google's ChromeBook;
  - Solaris, by Oracle (originally created by Sun Microsystems), a Unix-like OS most employed mostly in servers.
- A CS professional should be able to work using MacOS, Windows or Linux;
- A good developer should be able to develop to any OS!

Video: Most Popular OS timeline (all devices).

This Week's tasks

- Install VirtualBox in your machine (or create a Google Colab account) and share your experience in a post
- Programming project #7
- Quiz #3 covering topics 5, 6, and 7

Tasks

- Download **VirtualBox** to your machine and **install** it (or create a **Google Colab** account), then post about your experience.
- **Install a Linux Ubuntu virtual machine** or **create a Virtual Machine in Google Colab**.
- Quiz #3 available this Saturday

Post Discussion - Week 7 - In-class exercises

Preferred option: Download VirtualBox to your machine and install it, then post about your experience in the discussion forum.

Alternative option: Create an account at Google Colab to use a virtual machine remotely.

- Was it easy to install? Did it take too long?
- Was your machine particularly slow afterward?
- Did it itch you to install a first virtual machine on it?

Your task:

- Post your discussion in the message board until this Saturday;
- Reply posts of your colleagues in the message board until next Wednesday.

This task counts towards the In-Class Exercises grade.

Project #7 - Week 7 - VirtualBox - P#7

Install a Linux Ubuntu virtual machine in your VirtualBox, then install VSCode into your Linux, and run last week's coding assignment into it. Alternatively you can use another App, as Parallels too (or go for a VM in cloud option - see next slide).

- You have to download an ***Ubuntu*** version (feel free to try other OSs if you feel like) then install it in a new VM;
- Once installed, open a browser within it (***Firefox***) and download the appropriated ***VSCode*** distribution (.deb);
  - Just follow the steps in the VSCode page;
- Once you have ***VSCode*** installed make sure python is working (***python3*** command is already available at Ubuntu distribution);
- Having your VSCode operational for Python, run your Week 6 coding assignment program in there.

Project #7 - Week 7 - Google Colab - P#7

If for some reason you have an issue installing VirtualBox, you have the option to create an account at Google Colab and use this free account to run a Python program in it. Alternatively, you can use a different VM provider, like AWS).

- Go to **Google Colab** using a **Google credential** (for example your Merrimack account, or a personal @gmail.com account);
- Create a free account (or use an existing one you have already) to run a Python 3 program in it;
- Run your Week 6 coding assignment program in there.

MacOS users with a M1 or M2 hardware are likely to use this option, since VirtualBox installation may be challenging.

Project #7 - Week 7 - P#7

- You must submit print screens images (photos are ok too) showing that:
  - You have successfully create a VM (for Ubuntu or Colab);
  - VSCode is running on your Ubuntu VM (if you are not in the Colab option);
  - The output of a few runs of your Week 6 assignment program duly running on your Ubuntu VM or Google Colab VM.

Your task:

- Save your files and submit it in the appropriate Canvas until Next Wednesday.

A few useful links:

- VirtualBox test builds: https://www.virtualbox.org/wiki/Testbuilds
- How to Install MacOS on a VirtualBox VM: https://youtu.be/Lq8J-vFqH7w
- How to Enable & Install Hyper-V on Windows 11: https://youtu.be/kV67xY1HJb0
- Google Colab page: https://colab.research.google.com

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This assignment counts towards the Projects grade.

Quiz #3 - Available this Saturday

- This quiz covers the contents presented in Week 5, Week 6, and Week 7.
- The quiz is not timed, but the hard deadline to submit it is next Wednesday.
- The quiz is composed by 10 simple questions that you need to answer by choosing the correct answer.
- You can only submit the quiz once, so make sure you double checked your answers before submitting it.
- This quiz is open notes and you can consult any materials referenced in these slides to answer it, but the answer must be your own, not from others.

Your task:

- Submit your quiz in Canvas until Next Wednesday 11:59pm EST.

This assignment counts towards the Quizzes grade.

Week 7 of CSC 6303

- **Post in the discussions until Saturday, reply until next Wednesday**
- **Do the coding project until next Wednesday**
- **Take Quiz #3 until next Wednesday**
- **Next week the Final Exam**

Next Week - Multi-thread, parallelization, and scheduling